

AUXILIARY PROBLEMS

These are problems motivated by some of the [OUTSTANDING PROBLEMS](#). We have started this section in the hope that problems here, some of which are actually interesting in their own right, might generate some hints for attacking the relevant outstanding problems.

*(AUX1) (A. Myasnikov) (cf. Problem [\(O9\)](#)) Let m, n be positive integers, and H and K nontrivial subgroups of a free group such that $\text{rank}(H) = n$ and $\text{rank}(K) = m$. Which numbers between 1 and $(n-1)(m-1)$ can be realized as $\text{rank}(H \cap K) - 1$? In particular, can $(n-1)(m-1) - 1$ be realized?

[Background](#)

(AUX2) (A. Myasnikov) (cf. Problem [\(O9\)](#)) Is there an algorithm which for every pair of positive integers m, n determines whether or not the Hanna Neumann conjecture holds for all subgroups of ranks m, n in a free group F_2 ?

(AUX3) (A. Borovik, A. Lubotzky, A. Myasnikov) (cf. Problem [\(O1\)](#)) For a group G , by $d(G)$ we denote the minimal number of normal generators of G , and by $N_k(G)$, $k \geq d(G)$, the set of all k -tuples of elements of G which generate G as a normal subgroup. We say that a group G satisfies the *generalized Andrews-Curtis conjecture* if for any $k \geq \max(d(G), 2)$, two tuples $U, V \in N_k(G)$ are AC-equivalent in G if and only if their images are AC-equivalent in the abelianization of G .

(a) Find a group G which does not satisfy the generalized Andrews-Curtis conjecture.

(b) Does Grigorchuk's group satisfy the generalized Andrews-Curtis conjecture?

(AUX4) (A. Casson) (cf. Problem [\(O7\)](#)) Suppose ϕ is an onto homomorphism from F_{p+q} to $F_p \times F_q$. Is it true that $\text{Ker}(\phi)$ is the normal closure of precisely pq elements?

(AUX5) (S.V. Ivanov) (cf. Problem [\(O1\)](#))

(a) In addition to "elementary" operations on a set Y described in Problem (O1)(a), consider the following operation: replace some y_i by a free generator of the group F if this does not change the normal closure of Y . Is it true that every n -element set Y whose normal closure coincides with the whole $F = F_n$ can be reduced to X by a sequence of operations in Problem (O1)(a) together with this new operation?

(b) (weaker question) Similar question, but the ONLY operation is: replace some y_i by an element of the group F if this does not change the normal closure of Y . This problem is open if the cardinality of Y is greater than 2, and has the affirmative answer if it is equal to 2.

See [S.V. Ivanov, On balanced presentations of the trivial group, Invent. Math. 165 (2006), 525-549] for relevant discussions.